

Shunchang Liu

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RESEARCH INTERESTS

My research focuses on **AI Alignment**. I'm currently most interested in the underlying principles of alignment, i.e., understanding why models exhibit misaligned behaviors that deviate from human expectations, and how to control them from first principles. I've also worked on *robustness* and *copyright*.

EDUCATION

EPFL, Switzerland Sep 2026 — Present
PhD of Computer Science

EPFL - ETH Zürich (joint-degree program), Switzerland Sep 2024 — Aug 2026
Master of Computer Science: Cybersecurity GPA: 5.55 / 6.00 (Distinction)

Beihang University, China Sep 2017 — Jun 2021
Bachelor of Engineering: Automation GPA: 3.87 / 4.00 (Distinction)

RESEARCH EXPERIENCE

TML Lab, EPFL — PhD Student Sep 2026 — Present

SPY Lab, ETH Zürich — Master's Thesis Student Feb 2026 — Aug 2026

LAS Group, ETH Zürich — Semester Project Student Mar 2025 — Dec 2025

LIA, EPFL — Semester Project Student Sep 2024 — Jan 2025

Concordia AI — Intern Apr 2024 — Aug 2024, Jun 2025 — Aug 2025

Beihang University — Research Assistant Sep 2021 — Dec 2023

SELECTED RESEARCH PROJECTS

Emergent Misalignment in Vision-Language Models [paper] ETH Zürich
Master Thesis, Supervisor: Prof. Florian Tramèr and Prof. Francesco Croce Feb 2026 — Aug 2026

- Showed that narrow multimodal fine-tuning induces unsafe behavior on unrelated tasks, extending emergent misalignment from text-only to multimodal models.

Preference Instability Detection and Mitigation in Reward Models [paper] ETH Zürich
Semester Project, Supervisor: Prof. Andreas Krause and Dr. Xin Chen Mar 2025 — Dec 2025

- Found that RLHF reward models can reverse preferences under semantic-preserving perturbations, traced this instability to sparse internal features, and mitigated it without retraining the model.

Copyright Infringement Detection and Mitigation for Diffusion Models [paper] EPFL
Semester Project, Supervisor: Prof. Boi Faltings and Dr. Zhuan Shi Sep 2024 — Jan 2025

- Built CopyJudge, a VLM framework that assesses copyright infringement through multi-agent debate and reduces infringement risk in diffusion models through text-prompt and latent-noise optimization.

Adversarial Patch Generation for Crowd Counting [paper] Beihang University
Research Assistant, Supervisor: Prof. Aishan Liu and Dr. Yingwei Li (JHU) Jun 2021 — Aug 2022

- Designed scale- and position-aware adversarial patches that achieved SOTA attacks on crowd-counting models, improving robustness to complex backgrounds and cross-dataset generalization through adversarial training.

PUBLICATIONS

Co-first author (*) / Corresponding author (†)

Narrow Multimodal Fine-Tuning Can Induce Emergent Misalignment.

Shunchang Liu, Lukas Fluri, Xin Chen, Francesco Croce.

Preprint, 2026 [paper] [code]

Preference Instability in Reward Models: Detection and Mitigation via Sparse Autoencoders.

Shunchang Liu, Xin Chen, Belén Martín-Urcelay, Francesco Croce.

Preprint, 2026 [paper] [code]

CopyJudge: Automated Copyright Infringement Identification and Mitigation in Text-to-Image Diffusion Models.

Shunchang Liu*, Zhuan Shi*, Lingjuan Lyu, Yaochu Jin, Boi Faltings.

ACM Multimedia, 2025 [paper] [code]

Harnessing Perceptual Adversarial Patches for Crowd Counting.

Shunchang Liu*, Jiakai Wang*, Aishan Liu, Yingwei Li, Yijie Gao, Xianglong Liu, Dacheng Tao.

ACM CCS, 2022 [paper] [code]

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IP Protection in the Era of Visual Generative AI: A Survey.

Zhuan Shi, **Shunchang Liu**, Alireza Dehghanpour Farashah, Qian Yang, Han Yu, Cao Yang, Chaochao Chen, Yuping Yan, Yaochu Jin, Golnoosh Farnadi, Lingjuan Lyu.

Preprint, 2026 [paper]

The Biased Oracle: Assessing LLMs' Understandability and Empathy in Medical Diagnoses.

Jianzhou Yao*, **Shunchang Liu***, Guillaume Drui, Rikard Pettersson, Alessandro Blasimme, Sara Kijewski.

NeurIPS Workshop @GenAI4Health, 2025 [paper] [code]

Boosting Cross-task Transferability of Adversarial Patches with Visual Relations.

Tony Ma, Songze Li, Yisong Xiao, **Shunchang Liu†**.

IEEE CVPR Workshop @AdvCV, 2023 [paper]

Benchmarking the Robustness of Quantized Models.

Yisong Xiao, Tianyuan Zhang, **Shunchang Liu**, Haotong Qin.

IEEE CVPR Workshop @AdvCV, 2023 [paper]

Revisiting Audio Visual Scene-Aware Dialog.

Aishan Liu, Huiyuan Xie, Xianglong Liu, Zixin Yin, **Shunchang Liu**.

Neurocomputing, 2022 [paper]

Dual Attention Suppression Attack: Generate Adversarial Camouflage in Physical World.

Jiakai Wang, Aishan Liu, Zixin Yin, **Shunchang Liu**, Shiyu Tang, Xianglong Liu.

IEEE CVPR, 2021 (**Oral**) [paper] [code]

AWARDS

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| • EPFL EDIC PhD Fellowship | May 2026 |
| • Outstanding Graduate of Beihang University (Top 20%) | Jun 2021 |
| • CVPR Security AI Challenge: No-limit Adversarial Attacks on ImageNet (11/1559) | Apr 2021 |
| • Outstanding Student of Beihang University (Top 5%) | Jun 2019 |

SERVICES

- **Program Committee / Reviewer:** ICLR [2027], NeurIPS [2026], Pattern Recognition, Frontiers of Computer Science, IEEE TCSVT, IEEE T-ITS, and Workshops (ICML MoFA 2025, CVPR AdvCV 2023, etc.)
- **Teaching Assistant:** Machine Learning (EPFL Fall 2025)